EE206

2-STAGE OTA DESIGN

NAME: ADITYA SINGH

ROLL NO.: 240102115

Technology: 180 nm

Constraints:

$V_{DD} = 1.8V$

$V_{Tn} = 0.37V$

$V_{Tp} = 0.39V$

$\mu_n C_{ox} = 230\ \mu A/V^2$

$\mu_p C_{ox} = 100\ \mu A/V^2$

$W_{min} = 270\ nm$

$L_{min} = 180\ nm$

Gate Overdrive voltage = 200 mV

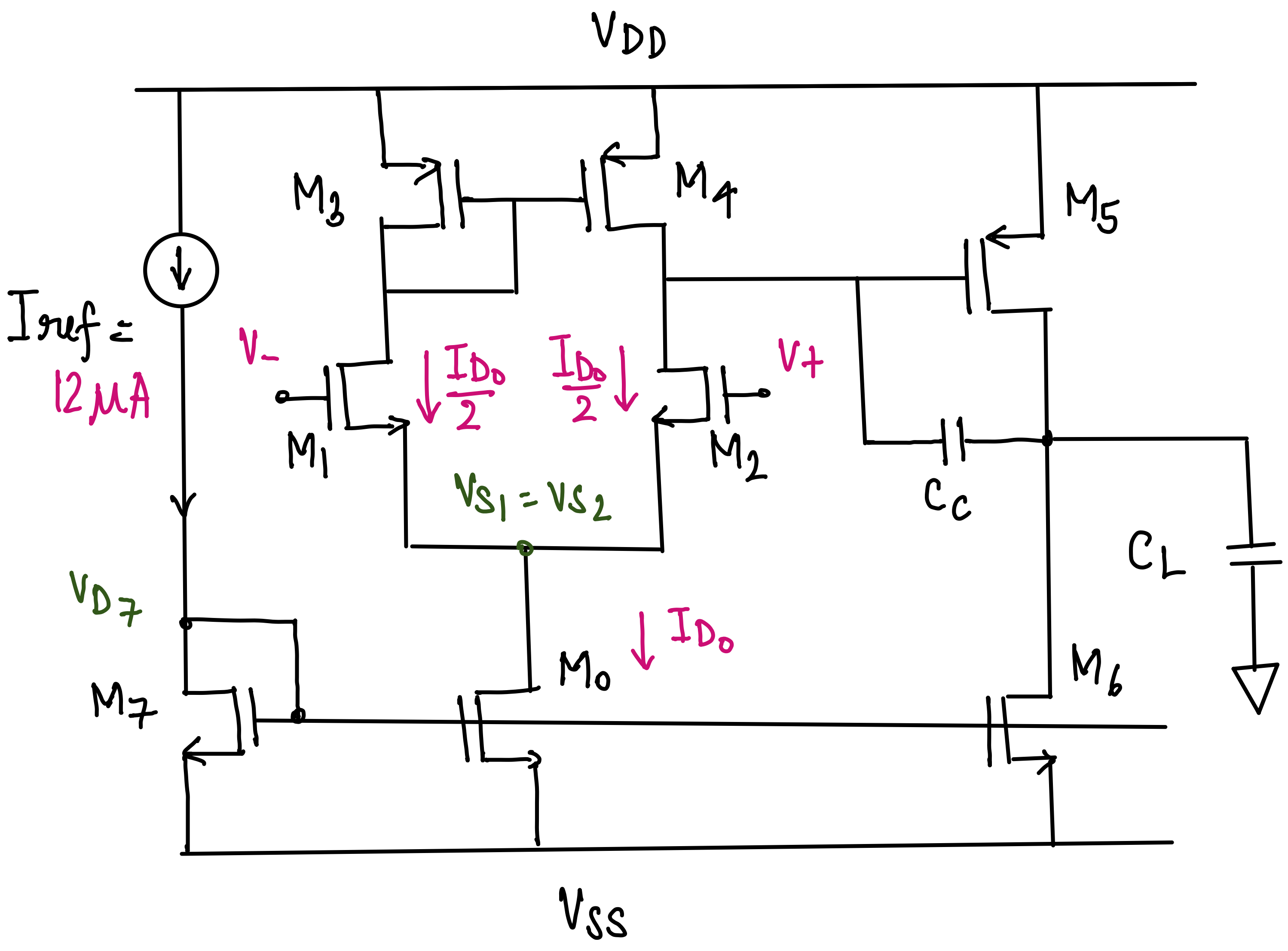
** for initial calculations

Requirements: DC Gain $\geq$ 40 dB

Closed Loop Gain = 2

Assumptions:

1. Value of reference constant current source = $12\ \mu A$.
2. Current in the second stage to be the same as the first stage.



M_0 : Tail Current source for first stage differential amplifier.

M_1, M_2 : Input differential pair

M_3, M_4 : Current Mirror Active Load.

M_5 : Second Stage Amplifier

M_6 : Second stage Current Source

M_7 : Current Mirroring transistor

CALCULATIONS

1. Reference current mirroring transistor M_7:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_7 (V_{GS} - V_{tn})^2$$

$$I_D = 12\,\mu A$$

$$\mu_n C_{ox} = 230\,\mu A/V^2$$

$$V_{gs} - V_{tn} = \Delta V = 0.2V$$

$$12 = \frac{1}{2} \times 230 \times \left(\frac{W}{L}\right)_7 \times (0.2)^2$$

$$\boxed{\Rightarrow \left(\frac{W}{L}\right)_7 = 2.6 \quad \{L = 1\,\mu m,\ W = 2.6\,\mu m\}}$$

2. Tail Current Source of Differential Pair M_0:

Given: $I_{ref} = \frac{1}{10} \times I_{D_0}$

$$\Rightarrow I_{D_0} = 10 \times 12 = 120\,\mu A$$

$$\left(\frac{W}{L}\right)_0 = \frac{2 I_{D_0}}{\mu_n C_{ox}(V_{GS} - V_{tn})^2} = 26$$

$$\boxed{\left(\frac{W}{L}\right)_0 = 26 \quad \{L = 1\,\mu m,\ W = 26\,\mu m\}}$$

3. Input Differential Pair M_1, M_2 :

Due to symmetry, through half circuit analysis.

$$I_{D_1} = I_{D_2} = \frac{I_{D0}}{2} = 60\,\mu A$$

$$I_{D_1} = I_{D_2} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{Th})^2$$

$$\boxed{\Rightarrow \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 13 \quad \{W = 13\,\mu m, \; L = 1\,\mu m\}}$$

4. Active Load Current Mirrors M_3, M_4 :

$$I_3 = I_4 = I_1 = I_2 = \frac{I_0}{2} = 60\,\mu A \quad ; \quad \mu_p C_{ox} = 100\,\mu A/V^2$$

$$I_3 = I_4 = \frac{1}{2} \times 100 \times \left(\frac{W}{L}\right)_3 \times (0.2)^2$$

$$\boxed{\left(\frac{W}{L}\right)_3 = \left(\frac{W}{L}\right)_4 = 30 \quad \{L = 1\,\mu m, \; W = 30\,\mu m\}}$$

5. Second Stage Amplifier

$$I_5 = I_1 = 120\,\mu A.$$

$$\Rightarrow I_5 = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_5 (V_{SG} - V_{Tp})^2$$

$$\Rightarrow \boxed{\left(\frac{W}{L}\right)_5 = 60 \quad \{L = 1\,\mu m, \; W = 50\,\mu m\}}$$

6. Second Stage Current Source

As stated in assumptions, we get $I_6 = I_0$

$$\boxed{\Rightarrow \left(\frac{W}{L}\right)_6 = \left(\frac{W}{L}\right)_0 = 26 \quad \{L = 1\,\mu m,\ W = 26\,\mu m\}}$$

For the circuit to function properly, we must ensure that every MOSFET is in saturation

Hence, we have to check for 2 conditions :

$V_{DS} > V_{GS} - V_{Tn}$ $\quad (V_{SD} > V_{SG} - V_{Tp})$

$V_{GS} > V_{Tn}$ $\quad (V_{SG} > V_{Tp})$

1) M_0 : $\quad V_{DS_0} = V_{S_1} - V_{SS}$. $\quad$ here, $V_{SS} = 0$.

$\Rightarrow V_{DS_1} = V_{D_1} - V_{S_1}$

$\Rightarrow V_{GS_1} - V_{Tn} = V_{G_3} - V_{S_1} - 0.37V = 0.2V$

$\Rightarrow 0.9 - V_{S_1} - 0.37V = 0.2V$

$\Rightarrow \boxed{V_{S_1} = 0.33\ V}$

$V_{GS_0} - V_{Tn} = V_{D_7} - 0 - 0.37 = 0.2V$

$\Rightarrow V_{D_7} = 0.57\,V.$

$V_{DS_0} = V_{S_1} - V_{SS} \Rightarrow V_{S_1} > 0.2$

$V_{GS} - V_{tn} > 0.37$

$\}$ $\therefore$ saturation ensured

2) M₁ :

$V_{DS1} = V_{D3} - V_{S1} = V_{D3} - 0.33$

$\Rightarrow V_{D3} > 0.53$

$V_{GS} - V_{Tn} = 0.9 - V_{S1} - 0.37 = 0.2$

satisfied as $V_{S1} = 0.33\ V$

3) M_2 : Same condition as M_1, in saturation as $V_X = 0.33V$

4) M_4: $V_{SD_4} = 1.8 - V_{D4}$

$V_{SG_4} - V_{Tp} = 1.8 - V_{G3} - 0.39 = 0.2$

$\Rightarrow V_{G3} = 1.21V$

$V_{D4} < 1.6V$

$V_{G3} < 1.41\ V$

saturation to be ensured. as $V_{G3} = 1.21V$

5) M_5 : $V_{SD5} = 1.8 - V_{out}$

$V_{SG5} - V_{Tp} = 1.8 - V_{D4} - 0.39 = 0.2$

$V_{out} < 1.6$

$V_{D4} < 1.41$

saturation ensured as $V_Z = 1.21V$

6) $\underline{M_6}$: $V_{DS6} = V_{out}$

$V_{GS} - V_{Th} = V_{G7} - 0.37 = 0.2$

$V_{G7} = 0.57$

$V_{out} > 0.2$

$V_{G7} > 0.37$

$\therefore$ saturation conditions satisfied.

7) M_3 and M_7 are always in saturation as they are diode connected.

We make sure $V_{out} \in (0.2V, 1.6V)$

Thus, every MOSFET will be in saturation

for maximum output swing, $V_{out} = 0.9V$

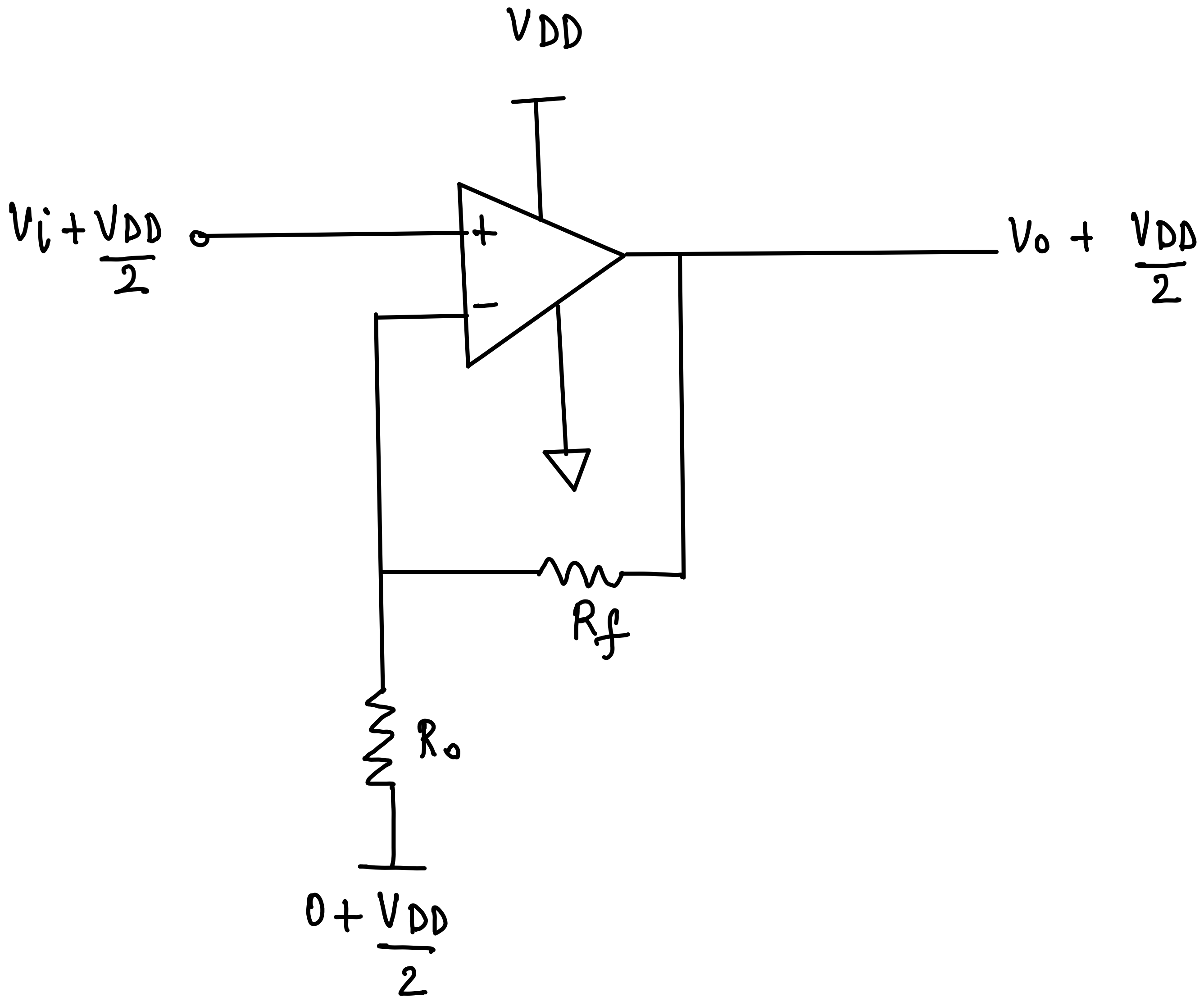
taking load capacitance $C_L = 10pF$.

for desired phase margin ($\approx 60°$), $C_C > 0.22 C_L$

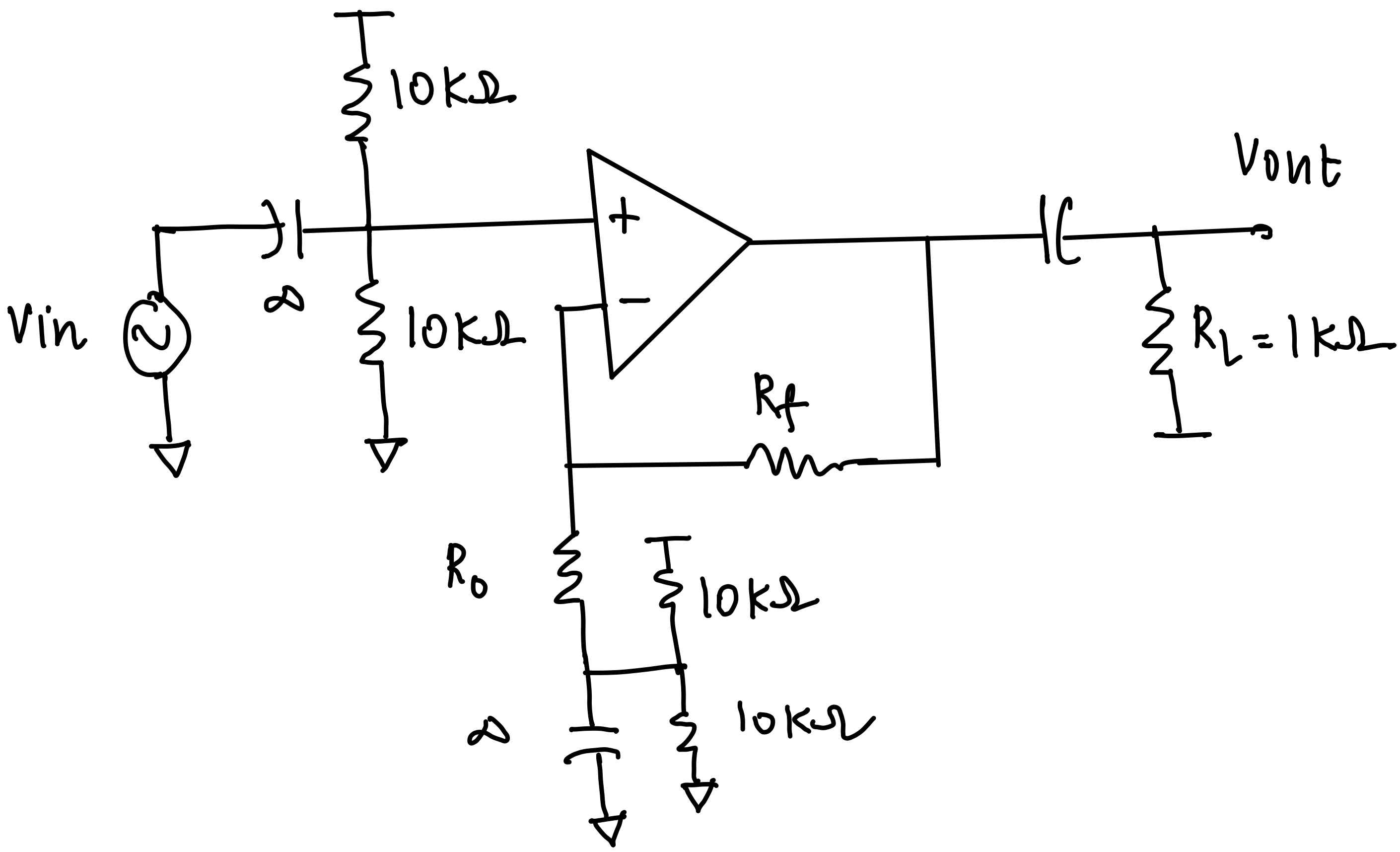
$\therefore C_C = 5pF$.

CLOSED LOOP NON-INVERTING AMP.

Gain = 2



AC signal must ride over $\frac{V_{DD}}{2}$ to ensure maximum swing limits



$$A = 1 + \frac{R_f}{R_0} = 2$$

$\therefore$ let $R_f = R_0 = 10K\Omega$

TABLE (from LTSPice)

MOS	W(μm)	L(μm)	gm (μA/V)	ro (in kΩ)	Vgs - Vtn (V)	ID (μA)
M_0	26	1	1248.5	88.5		112.96
M_1	13	1	624.5	177.1		56.46
M_2	13	1	624.3	177.0		56.46
M_3	30	1	531.3	212.5		56.46
M_4	30	1	531.5	212.4		56.46
M_5	60	1	1099.4	99.3		120.88
M_6	26	1	1291.6	82.7		120.88
M_7	2.6	1	128.7	833.3		12.00

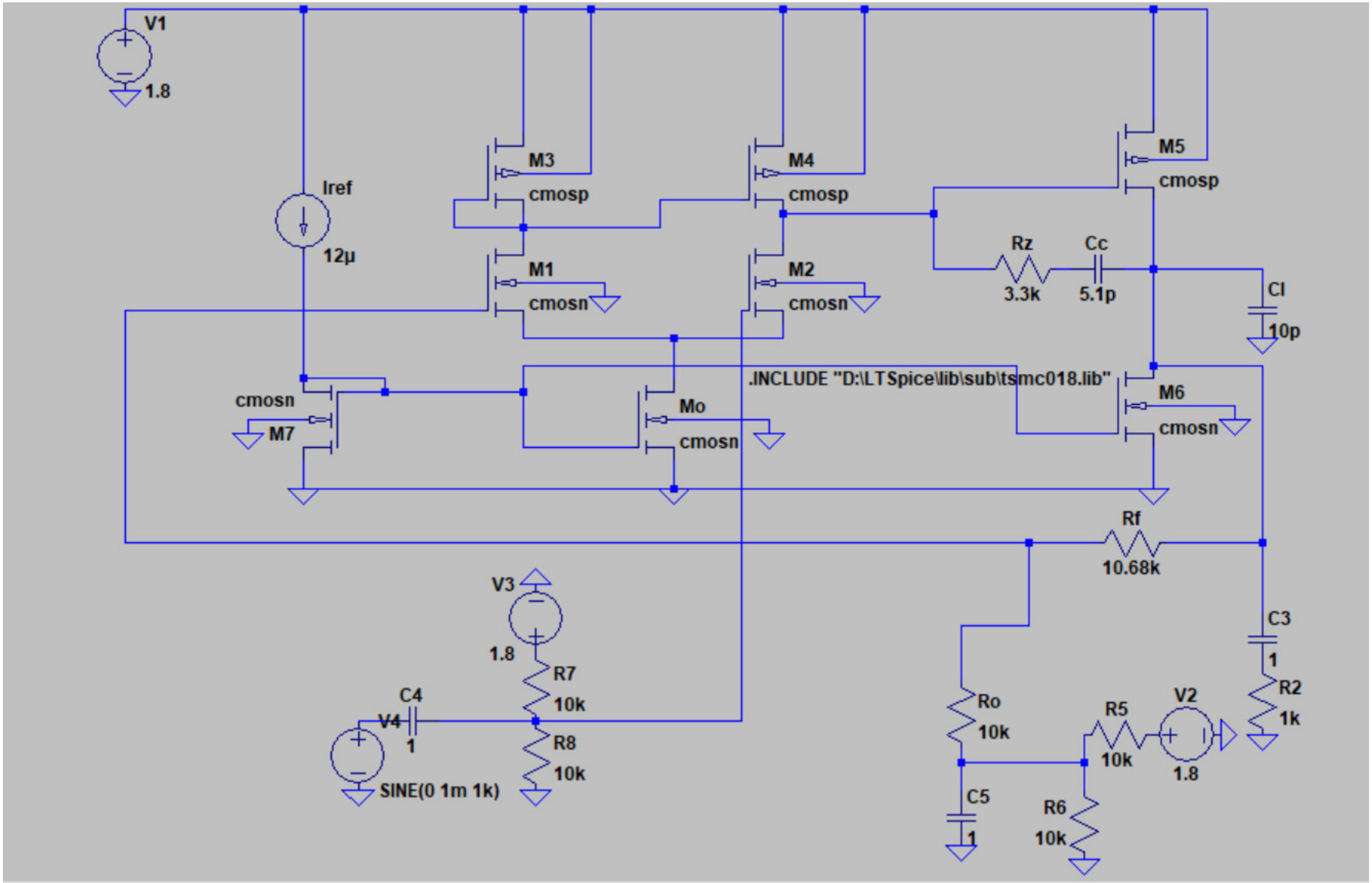
Computed Gain = 69.5 dB > 40 dB
total linear gain = 2991 V/V
power consumption = 442.5 μW

I_{M_0} calculated = 120 μA
actual = 112. μA } pretty close

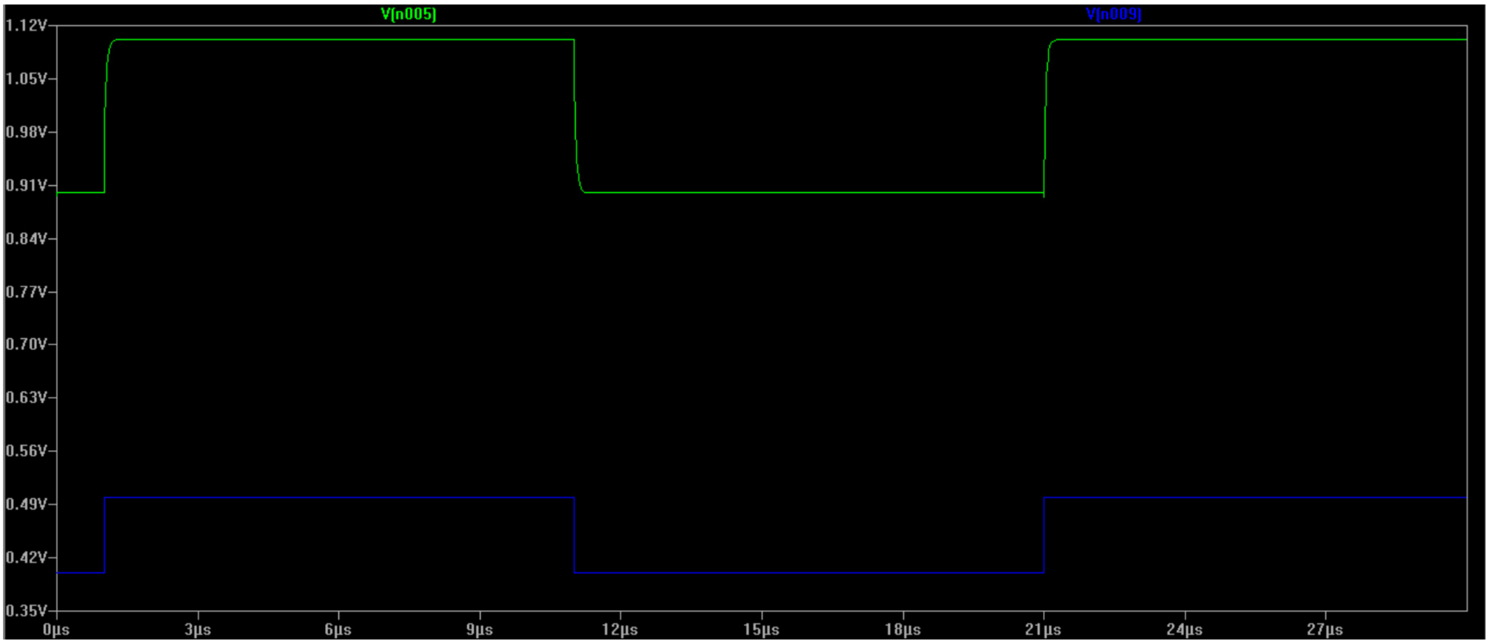
$I_{D1} = I_{D2} = I_{D3} = I_{D4}$

$I_{D0} \approx I_{D7} \times 10 = I_{D6}$

FULL CIRCUIT SCHEMATICS IN NEXT PAGE


V1
1.8
Iref
12µ
M3
cmosp
M4
cmosp
M5
cmosp
M1
cmosn
M2
cmosn
Rz
3.3k
Cc
5.1p
Cl
10p
cmosn
M7
Mo
cmosn
.INCLUDE "D:\LTSpice\lib\sub\tsmc018.lib"
M6
cmosn
Rf
10.68k
V3
1.8
R7
10k
C4
V4
1
R8
10k
SINE(0 1m 1k)
C3
1
R2
1k
Ro
10k
R5
10k
V2
1.8
C5
1
R6
10k

CLOSED LOOP OUTPUT (A=2)



OPEN LOOP AC ANALYSIS